Johnna Szkudlarek

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Education

Harvard College — Cambridge, MA
Bachelor of Science in Mechanical Engineering — Expected May 2028

Work Experience

D'Angelo Technologies LLC — Xenia, OH
Mechanical Engineering Intern — June 2026 – Present

- Redesigned an existing CAD model in SolidWorks and created laser-cut graphic overlays using LightBurn, producing low-cost, visually accurate training units within one week.
- Designed and 3D printed a mechanical linkage connecting a vehicle gear selector to a potentiometer.
- Improved durability of 3D-printed controller button caps by replacing fragile snap fit features with a press fit design, improving button retention.
- Assembled and wired 25 custom controllers with 250 total multi-configuration buttons, utilizing Lean Six Sigma principles to optimize cutting, routing, and soldering processes while ensuring consistent, high-quality production.
- Created detailed engineering drawings documenting button placement, switch types, and assembly configurations, enabling other technicians to assemble the controllers accurately and consistently.

SOCHE — Brookville, OH
Technology Support Summer Intern — May 2023 – Aug 2023

- Reimaged, configured, and troubleshot 200+ workstations and classroom/office technology systems.
- Reassigned 700+ Chromebooks and chargers to support district-wide student distribution.

Engineering Experience

Harvard Undergraduate Automotive Society — Cambridge, MA
Manufacturing Lead — Jan 2025 – Present

- Redesigned the pedal box for an internal combustion race car.
- TIG welded steel tabs onto the chassis to support rear suspension and engine installation.
- Researched and designed a heat shield in compliance with Formula SAE rules.

Harvard Undergraduate Rover Club – Chassis Project — Cambridge, MA
Welding and Design Engineer — May 2025 – Aug 2025

- Fabricated a 44" × 21.68" × 12" chassis from 1" × 1" × 1/16" aluminum square tubing using TIG welding.
- Oversaw design choices with materials and ensured compliance with the University Rover Competition 2026 rules.

Turf War Robot — Cambridge, MA
Team Project — Aug 2024 – Dec 2024

- Designed and manufactured a 9.5" × 9.5" × 5" competition robot as part of a 5-person team.
- Used SolidWorks CAD, laser-cut acrylic, 3D printing, lathe, and mill during prototyping and fabrication.
- Earned Second Place in the class Turf War Robot Competition.

Plywood Canoe — Cambridge, MA
Team Project — Aug 2024 – Dec 2024

- Built a 14' × 4' plywood canoe in a 4-person team and successfully tested it on the Charles River.
- Gained hands-on fabrication experience using jigsaws, drills, sanders, and epoxy application techniques.

Technical Skills

CAD/Engineering: SolidWorks
Fabrication: TIG and MIG welding, 3D printing, lathe, mill, laser cutting, hand and power tools
Languages: Spanish (Proficient)